Lid Muscle Abnormalities

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Lecture Acknowledgement and Thanks:

Professor William C. Hefner

Pacific University



SOAP Format

- Subjective
- Objective
- Assessment
- Plan

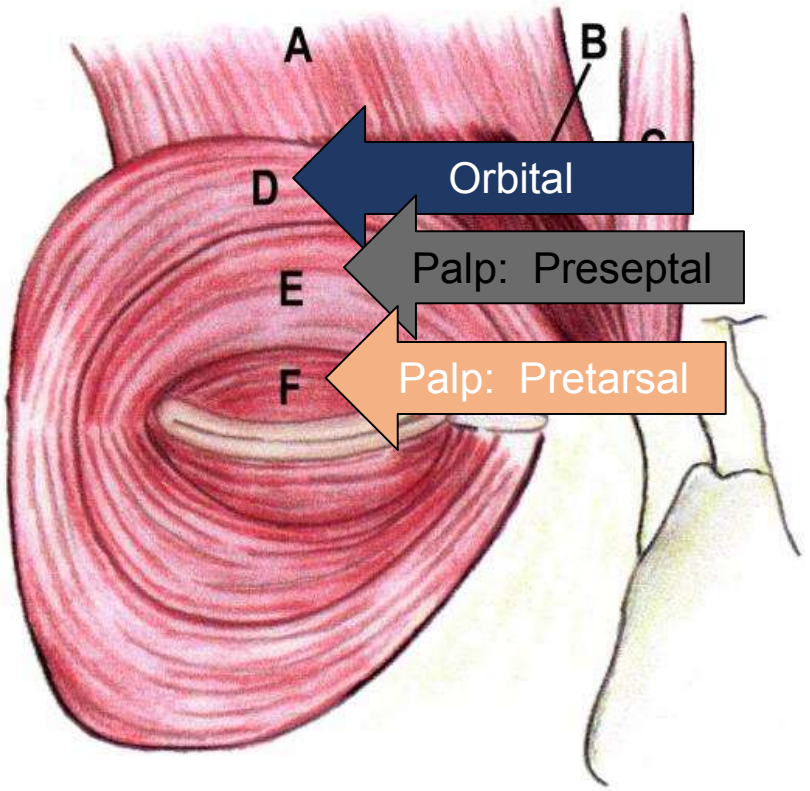


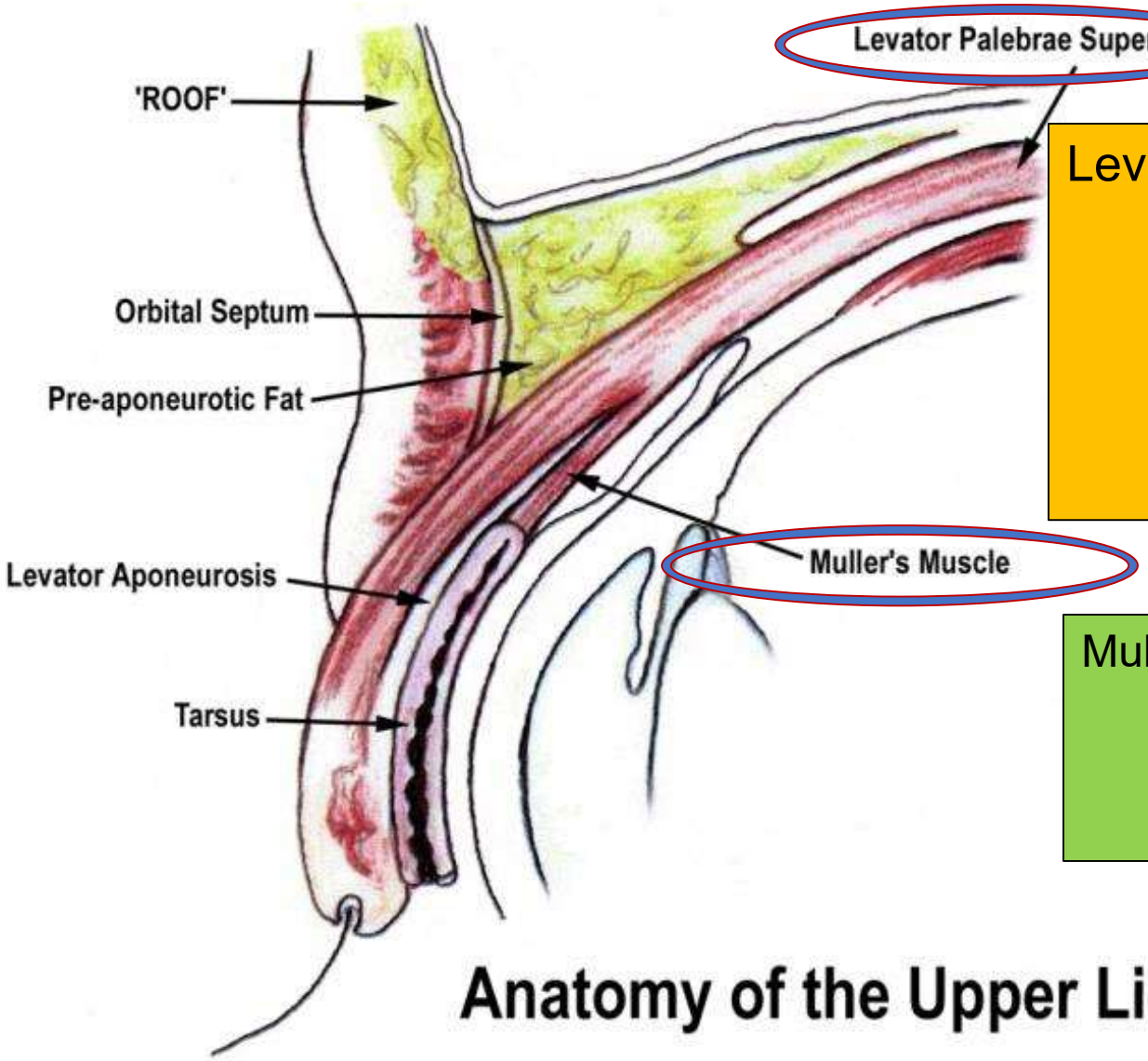
Eyelid basics

- Interpalpaebral Fissure Width (Height) (aka PFH):
  - Young adults 10-11 mm
  - Older adults 8-10 mm
- Upper lid margin 1-2 mm BELOW s. limbus
- Tarsal plates are ~25 mm horizontally
  - Meibomian glands housed within
    - ~25 in upper lid
    - ~20 in lower lid

Eyelid basics- the muscles

- Orbicularis oculi
  - Orbital portion
  - Palpaebral portion
    - Pre-Septal
    - Pre-Tarsal
    - Ciliary (aka Riolan Muscle)
  - Lacrimal portion
- Levator Palpebrae
- Muller's





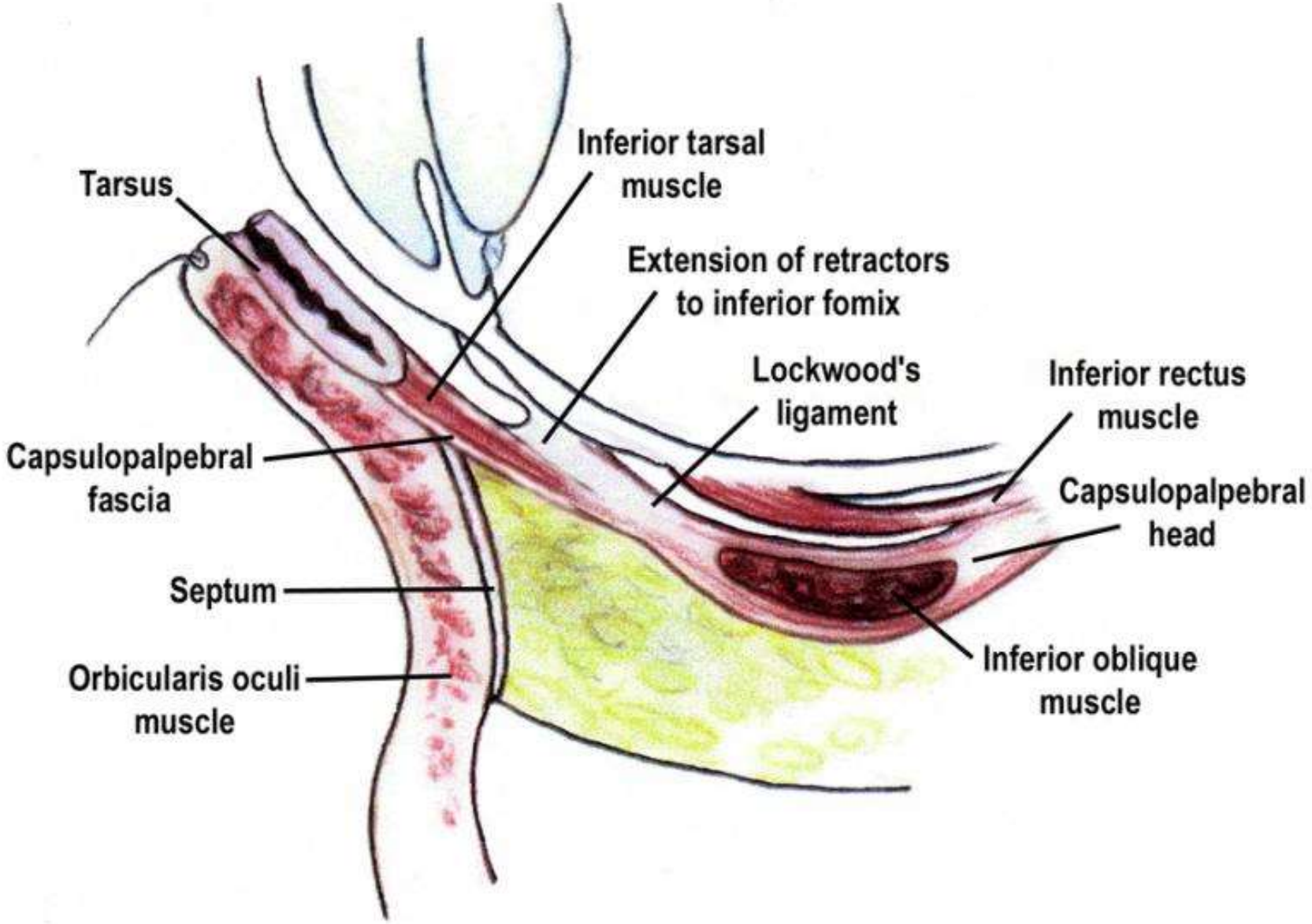
Levator:

- Skeletal muscle
- Opening the eye
- Innervation is superior branch of CN III
- Raises lid ~15 mm

Muller's:

- Smooth muscle
- Innervation is CN III
- Raises lid ~1-2 mm

http://emedicine.medscape.com/article/834932-overview#showall



Anatomy of the Lower Lid

http://emedicine.medscape.com/article/834932-overview#showall

Lid position determined by:

1. Tone of Orbicularis Oculi Muscle
   closes the eye
2. Tone of Levator Palpaebrae Muscle
   opens the eye



s this ma



Normal?

Abnormal?



http://webeye.ophth.uiowa.edu/eyeforum/tutorials/ptosis/2a-LRG-CN3.jpg

Roles of the lids

- Protection
- Tears:
  - Production
  - Distribution
  - Removal
- Cosmesis

What do you think the patient is experiencing? (S) ptosis → diagnosis



https://www.seattleface.com/service/face-and-neck-surgery/ptosis-repair

What is the problem here? (O)

ptosis



https://www.seattleface.com/service/face-and-neck-surgery/ptosis-repair

What do you think caused the problem? (A)



https://www.seattleface.com/service/face-and-neck-surgery/ptosis-repair

How do you want to fix this? (P)

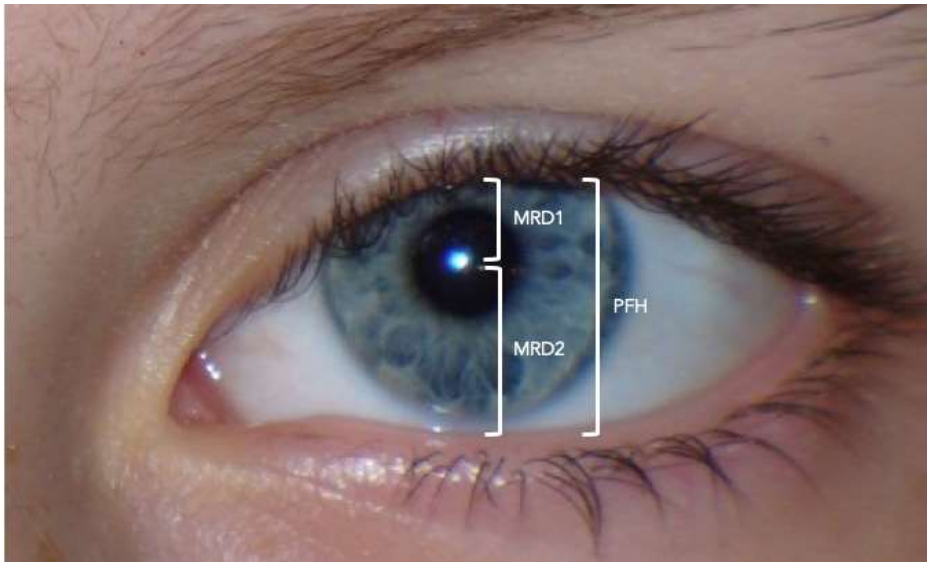
surgical



https://www.seattleface.com/service/face-and-neck-surgery/ptosis-repair

Ptosis: Abnormally low position (drooping) of upper eyelid

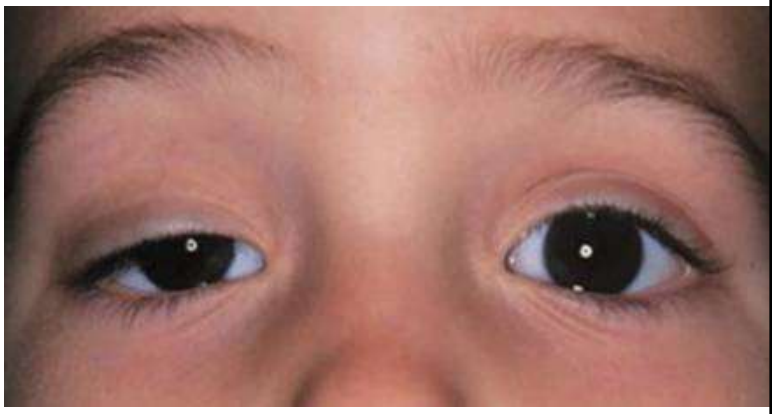




- Also called blepharoptosis

- Types
  - Congenital (born with it)
  - Acquired (later in life)

Congenital Ptosis



- Present at birth
- 90% ptosis cases
- 70% unilateral
- 30% have strabismus
- 12% have anisometropia >1.00D
- 20% have amblyopia
  - Typically deprivational

Congenital Ptosis

Causes

1. Myogenic (Irregularity of levator muscle)
2. Neurogenic (Abnormality of CN III superior division)
3. Anomalous Innervation (Marcus Gunn Jaw Winking)

blinking when eating






Congenital Ptosis

- Myogenic (Irregularity of the levator muscle)
  - Majority of congenital ptosis cases
  - Present at birth, stable throughout life
  - Upper eyelid crease normally ABSENT

Lid position in downgaze should be noted

  - In myogenic-caused congenital ptosis, ptotic lid appears higher in downgaze
  - In acquired ptosis, the ptotic lid is equal to the non-ptotic lid



Congenital Ptosis

- Neurogenic (Abnormality of CN III superior branch)
  - Disrupted innervation to Muller's muscle
  - Most common occurrence is Congenital Horner's
    - Ptosis, miosis, anhidrosis and often iris heterochromia
    - Look for Harlequin Sign (flush of unaffected side of face)



Morrison DA, Bibby K, Woodruff G. The "harlequin sign and congenital Horner's syndrome. *J Neurol Neurosurg Psychiatry. 1997;*62:626-628

Congenital Ptosis

- Anomalous Innervation
  - Marcus Gunn "jaw-winking" syndrome
    - Chewing/moving jaw to opposite side → lid elevation
    - Misdirection of either:
      - Efferent motor innervation
      - Afferent proprioceptive fibers of CN III and V



https://collections.lib.utah.edu/details?id=188578&q=eyelid+disorder

Acquired Ptosis

- Results from:
  - Affliction of nerve supply to lid musculature
  - Diseases of lid muscles
  - Mechanical interferences in elevation

Acquired Ptosis



- Causes:
  - Neurogenic
  - Myogenic
  - Aponeurotic
    - Where levator aponeurosis has undergone structural changes which reduces its ability to lift eyelid
  - Mechanical

Acquired Ptosis

- Neurogenic
  - Horner's syndrome
    - Ipsilateral findings of:
      - mild ptosis
      - miosis
      - anhidrosis

↳ lack of sweating



https://www.semanticscholar.org/paper/Orbital-rhabdomyosarcoma.-0ffd0197821f76

https://openi.nlm.nih.gov/detailedresult.php?img=PMC3086325_SHORTS-10-10101&req=4

Acquired Ptosis neurological check-up



https://www.semanticscholar.org/paper/Orbital-rhabdomyosarcoma.-Karcioglu-Hadjistilianou/b3f8f8b31ff83e2f92b9e5f2ad0ffd0197821f76

Acquired Ptosis

- Neurogenic
  - Deficient innervation of CN III to levator
  - Most common associated conditions:
    - Diabetes mellitus (DM)
      - "Down and out" affected eye (CN III)
    - Horner's Syndrome (ear hurting!)
      - "Ptosis, miosis, anhidrosis" affected side
    - Aneurysms of internal carotid artery

Acquired Ptosis (Neurogenic)



Defective Adduction



Defective Elevation



Defective Depression



Normal Abduction

Acquired Ptosis

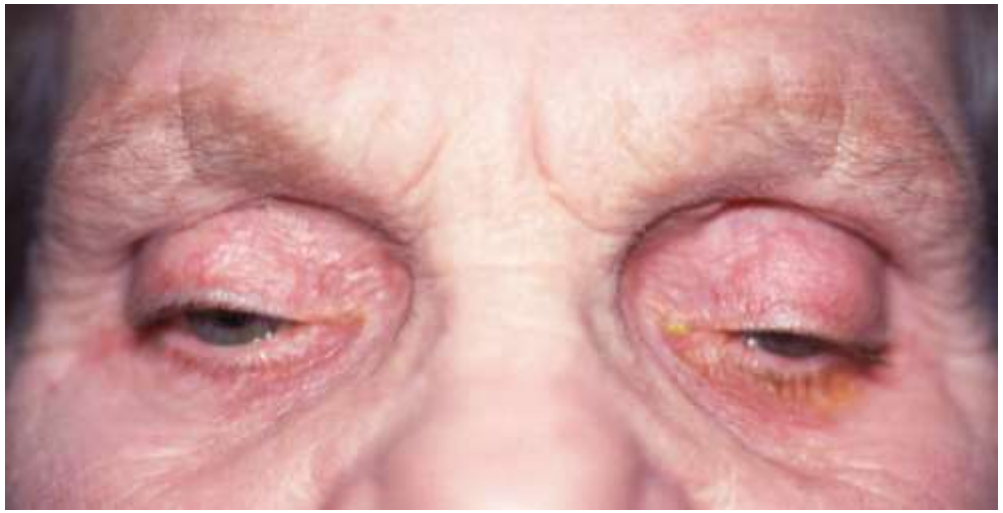
- cold pack can diagnose it

- Myogenic
  - Myasthenia Gravis
    - Abnormality at neuromuscular junction
    - Fatigability of voluntary muscle (Tensilon, cold pack test)
    - 96% have ptosis: bilateral but asymmetrical
    - Pt may have history of fluctuating ptosis & strabismus



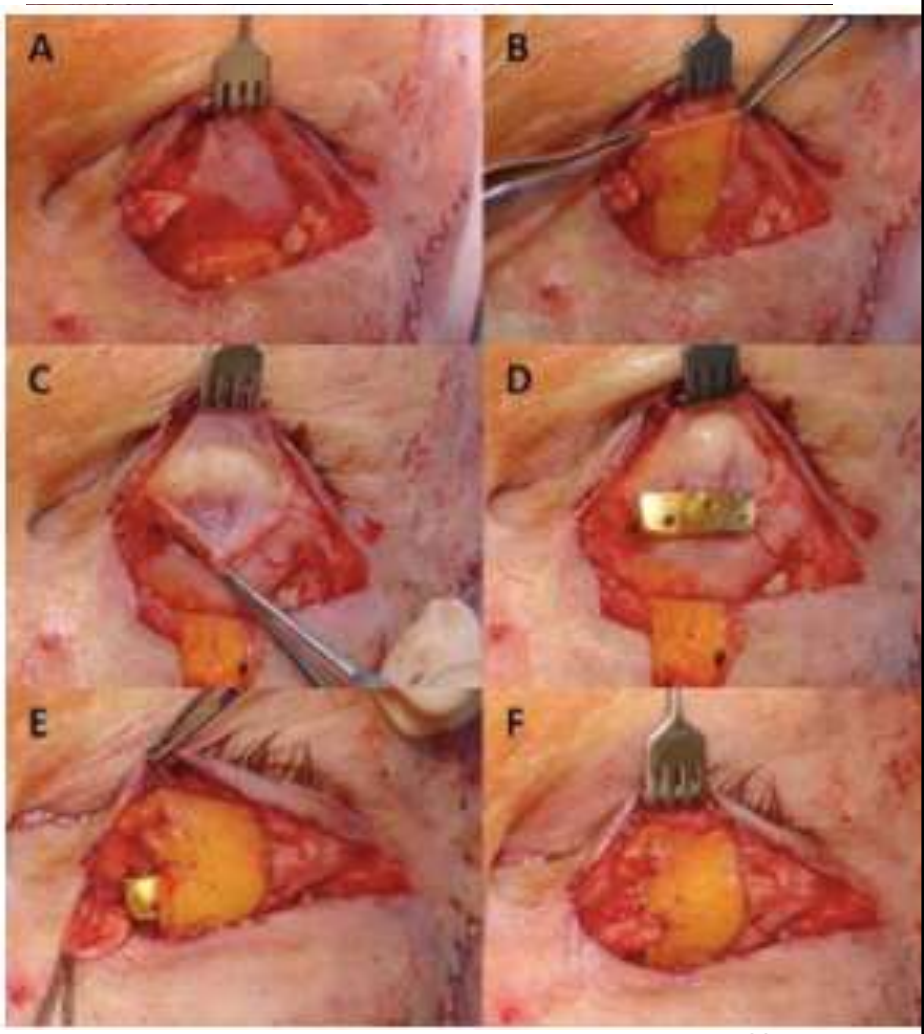
Acquired Ptosis

- Aponeurotic Dehiscence
  - Loss of levator tone and/or degeneration of levator attachments
    - Usually bilateral
  - Most common acquired ptosis



Acquired Ptosis

- Mechanical
  - Excessive weight
    - Edema
    - Tumors
    - Eyelid plates
  - Traumatic
    - Lacerations effecting nerves or muscles



Ptosis vs. Pseudoptosis

Looks like ptosis, but is not

- Upper lid looks low, lid retraction is normal

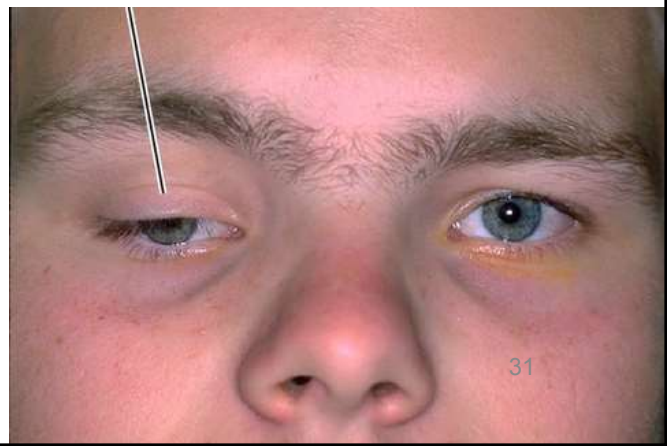


- Pseudo-ptosis conditions:
  - Dermatochalasis
  - Enophthalmos



  - Hypotropia — one eye lower than the other
  - Prosthetic eye





Ptosis

- Clinical Evaluation
  - HX what brings you in today
    - Onset: When first noticed?
    - FMHx: Anyone in family with ptosis?
    - Diplopia: Double vision?
    - Variability: Does it change throughout day?
    - Systemic symptoms/conditions
    - Trauma history
    - Allergic or drug reaction history – may cause edema

Ptosis

- Clinical Evaluation
  - Objective
    - Ptosis amount
    - Levator function
    - Pupils

Ptosis

- Clinical Evaluation (continued)
  - Ptosis amount
    - Measure both palpebral fissures and note the difference

Severity	Difference between fissures
Mild	1 – 2mm
Moderate	2 – 3 mm
Severe	≥ 4 mm

Ptosis

- Clinical Evaluation (continued)
  - Levator Function

Place ruler with 1mm marking at upper lid margin in downgaze



Normal levator function is 15 mm

Have patient look up. Difference between the downgaze & upgaze = function



Ptosis



- Clinical Evaluation (continued)
  - If congenital ptosis obscures any part of pediatric patient's visual field, surgery must be performed to correct problem early in life
    - Otherwise, permanent vision loss may occur as result of amblyopia:
      - Occlusion (deprivational) amblyopia
      - Astigmatism from compression by droopy eyelid

Ptosis

Differentials:

lack of movement of the eye
Eye "frozen"

- Chronic Progressive External Ophthalmoplegia
- Myasthenia Gravis
- Papilledema swollen
- Acquired myogenic ptosis
- Aponeurotic ptosis
- Neurogenic ptosis
- Mechanical ptosis
- Traumatic ptosis
- Pseudoptosis
- Blepharochalasis excess tissue
- Myotonic dystrophy
- Kearns-Sayre syndrome
- Acquired blepharoptosis

Ptosis

- Non-Surgical Treatment
  - Mild Cases
    - Observation if no signs of amblyopia, strabismus, and abnormal head posture

Ptosis

- Non-Surgical Treatment (continued)
  - Mild cases



www.pediatriceyemd.com

  - Follow-up every 3-4 months for signs of amblyopia due to congenital ptosis

Ptosis

- Non-Surgical Treatment (continued)
  - Head posture should be carefully examined
    - If patient acquires a chin-up posture due to worsening of ptosis, surgery may be indicated
  - Patient should be checked for astigmatism due to compression by droopy eyelid

Ptosis

- Surgical Treatment
  - Functional reasons
  - Physical reasons
  - Psychological reasons

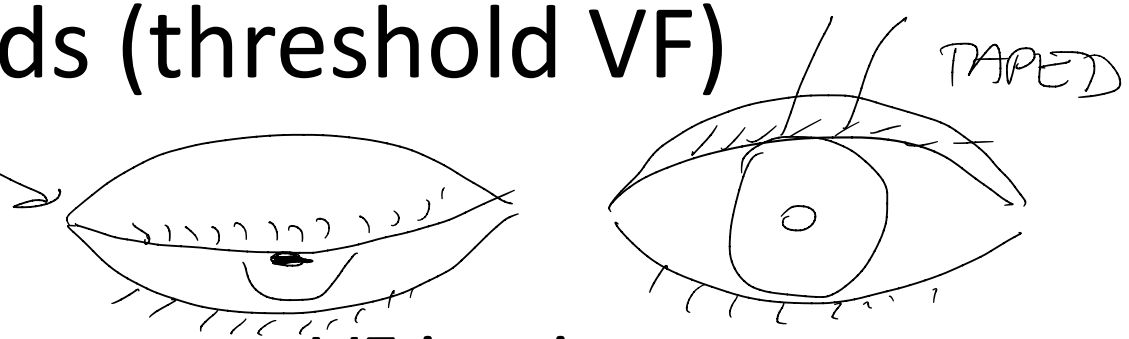
(self awear - cosmetical)



www.tedmontgomery.com

Ptosis

- Pre-Surgical Treatment should include:
  - VA's
  - Visual fields (threshold VF)
    - Untaped
    - Taped
    - Must demonstrate VF impingement



surtent improvement

Ptosis

- Surgical Treatment
  - Levator shortening
    - Must have some levator function
  - Frontalis suspension procedure
    - Levator function must be poor
    - The lids are linked to frontalis movement



www.youreyesbydrmary.com

Looks supried, wrinkeld forehead





https://www.seattleface.com/service/face-and-neck-surgery/ptosis-repair

Ptosis summary

- Lid droop
- Congenital
- Acquired
  - Multiple types
- Careful history and follow-up
- Treatment ranges from reassurance to surgery

Blinking

- Average rate 15-20 blinks/minute
- One blink takes about 0.25 seconds
- Disease may alter rates (ex. Parkinson's, Hyperthyroid)
- Partial blinks not coplimlit blink → very common
- Aggravated by keratoconjunctivitis sicca (dry eye)
  - Remember contact lens wearers

Blepharoclonus

- Increased frequency of blinking (children more commonly)
- Increased closure phase
- Caused by
  - Irritation
  - Inflammation
- Most frequently seen in children

Myokymia - trzanje

- Orbicularis fasciculation
- “An annoying twitch”

- Etiology:
  - Stress, tension, lack of sleep, caffeine, etc.

Myokymia

- Therapy:
  - Remove cause
  - Botulinum toxin type A (Botox)
    - Injected into affected area
    - 2.5-5U IM q3mo

Myokymia



- Treatment (continued)
  - Antihistamines
    - Antazoline (Vasocon A) q4h
    - Pheniramine (Opcon A, Naphcon A, Visine A) q4h
    - Promethazine (Phenergan) 12.5-25mg PO qd-tid
  - Quinine 200-300mg PO qd-bid
    - Tonic Water!!!!! helps relax the muscle

Blepharospasm



- Definition: Any abnormal tic or twitch of eyelid
- Bilateral involuntary orbicularis contraction: results in eye closure
- Spasmodic

no reason for it

- Idiopathic

Blepharospasm



- Patients over 50
- F>M
- There are at least 50,000 cases of blepharospasm in United States
  - Up to 2000 new cases diagnosed annually
  - Prevalence approximately 5 in 100,000

Blepharospasm

- Subjective
  - At one end of spectrum - essential blepharospasm - manifested by simple increased blink rate and intermittent eyelid spasms
  - On the other end of the spectrum - blepharospasm is disabling condition with ocular pain and functional blindness
  - Blepharospasm is commonly associated with dystonic movements of other facial muscles

Blepharospasm

- Subjective (continued)
  - Changes associated with long-standing blepharospasm include:
    - Eyelid and brow ptosis
    - Dermatochalasis
    - Entropion
    - Canthal tendon abnormalities

Blepharospasm

- Early symptoms of blepharospasm include:
  - Increased blink rate (77%)
  - Eyelid spasms (66%)
  - Eye irritation (55%)
  - Midfacial or lower facial spasm (59%)
  - Brow spasm (24%)
  - Eyelid tic (22%)

Blepharospasm

- Methods to relieve blepharospasm include:
  - Sleep (75%)
  - Relaxation (55%)
  - Inferior gaze (27%)
  - Artificial tears (24%)
  - Traction on eyelids (22%)
  - Talking (22%)
  - Singing (20%)
  - Humming (19%)

Blepharospasm

- Comorbid diagnoses include:
  - Dry eyes (49%)
  - Other neurologic disease (8%)
  - Video: http://www.kaltura.com/index.php/extwidget/preview/partner_id/797802/uiconf_id/27472092/entry_id/0_ts42qgoe/embed/auto?

Blepharospasm

- Differential Diagnoses:
  - Secondary to:
    - Foreign body (FB)
    - Trichiasis
    - Keratitis
  - Meige's syndrome
  - Hemifacial spasms
  - Facial Myokymia
  - Eyelid Myokymia
  - Facial Tic



Blepharospasm

- Therapy
  - Reassurance!
  - Oral meds are not too effective
  - Botulinum toxin type A (Botox)
    - Injected along upper and lower eyelid
    - 20-25U per eye IM q3mo
    - Most require re-injection in 3-4 months

Blepharospasm

- Therapy (continued)
  - Surgical
    - Last choice in therapy
    - Involves removal of orbicularis
    - Reserved for those unresponsive to botulinum toxin

Blepharospasm DDx

Differential Diagnosis	Gender	Age	Laterality	Voluntary Control	Clinical Features
Essential Blepharospasm	**F>M**	**>50**	**Bilateral**	**No**	**Isolated spasms of orbicularis**
Meige's Syndrome	**F>M**	**>50**	**Bilateral**	**No**	**Blepharospasms and midfacial spasms**
Hemifacial Spasm	**F>M**	**>45**	**Unilateral**	**No**	**Spasms along the distribution of CNVII**
Facial Myokymia	**F=M**	**Any**	**Unilateral**	**No**	**Rapid muscle flickering**
Eyelid Myokymia	**F=M**	**Any**	**Unilateral**	**No**	**Fasciculations of the orbicularis**
Facial Tic	**F=M**	**Youth**	**Bilateral or Unilateral**	**Yes**	**Repetitive movements**

Video Resource

Link to multiple videos of ocular and other neurological problems:

http://content.lib.utah.edu/cdm4/browse.php?CISOROOT=/ehsl-shw

THANK YOU!